

Defuzzification

Lambda cut

Lambda cut of a fuzzy set A is denoted as A_λ where $0 \leq \lambda \leq 1$.

It is a crisp set of all elements which have membership values greater than or equal to λ .

$$A_\lambda = \{x \mid \mu_A(x) \geq \lambda\}$$

Properties of lambda cuts

1. $(A \cup B)_\lambda = A_\lambda \cup B_\lambda$
2. $(A \cap B)_\lambda = A_\lambda \cap B_\lambda$
3. $\overline{A}_\lambda \neq \overline{A_\lambda}$ except when $\lambda = 0.5$.
4. If $\alpha \geq \lambda$ and $0 \leq \alpha \leq 1$, then $A_\alpha \subseteq A_\lambda$.

Membership

Membership features, i.e. core, support and boundary, can be denoted by lambda cuts.

1. $\text{core}_A = A_1$
2. $\text{supp}_A = A_{0+}$
3. $\text{bound}_A = A_{0+} - A_1$

Lambda cut for fuzzy relations

Lambda cut for a fuzzy relation R can be defined similar to that of fuzzy sets.

$$R_\lambda = \{(x, y) \mid \mu_R(x, y) \geq \lambda\}$$

Nearest ordinary set

Any λ -cut of a fuzzy set A is called a nearest ordinary set of A .

Note that there can be an infinite number of nearest ordinary sets of A .

Index of fuzziness

The difference of a fuzzy set A from its nearest ordinary set is a measure of the index of fuzziness of A , $\nu(A)$.

$$\nu(A) = \frac{2}{n^k} \cdot d(A, A_\lambda)$$

In the case of Hamming distance,

$$\nu(A) = \frac{2}{n} \cdot \sum_{x \in X} |A(x) - A_\lambda(x)|$$

Defuzzification

Defuzzification is the conversion of a fuzzy quantity to a precise quantity.

Defuzzification methods

1. Max membership method

For a fuzzy set A defined on universe X , the defuzzified value x^* is the element with the maximum membership value, i.e.

$$x^* = x \mid \mu_A(x) = h(A)$$

Simplest method.

2. Mean max method

For a fuzzy set A defined on universe X , the defuzzified value x^* is the mean of the elements with the maximum membership value, i.e.

$$x^* = \text{mean}(x_1, x_2, \dots, x_k) \mid \mu_A(x_i) = h(A) \quad \forall 1 \leq i \leq k$$

Similar to max membership method. Applicable when there are plateau or multiple peaks.

3. Centroid method

For a fuzzy set A defined on universe X , the defuzzified value x^* is described by the equation:

$$x^* = \frac{\int x \cdot \mu_A(x) dx}{\int \mu_A(x) dx}$$

Most accurate, but involves intensive calculation.

4. Weighted average method

For a fuzzy set A defined on universe X , the defuzzified value x^* is described by the equation:

$$x^* = \frac{\sum_{x \in X} x \cdot \mu_A(x)}{\sum_{x \in X} \mu_A(x)}$$

Less computationally extensive than centroid method. There is some geometric approximation involved here, so it is applicable for symmetric membership functions.

5. Center of sums method

If there is a union of fuzzy sets $A_1, A_2, \dots, A_n$, then the centroid of each, then we can find x^* from the equation:

$$\begin{aligned} x^* &= \frac{\int \bar{x} \cdot \sum_{k=1}^n \mu_{A_k}(x) dx}{\int \sum_{k=1}^n \mu_{A_k}(x) dx} \\ &= \frac{\sum_{k=1}^n \bar{x} \cdot \int \mu_{A_k}(x) dx}{\sum_{k=1}^n \int \mu_{A_k}(x) dx} \end{aligned}$$

In this method, the intersecting regions are calculated twice, which may cause some deflection from the result obtained using centroid method.

6. Center of highest area

In this method, only the membership function with the highest area, say A_k , is chosen. Its center is considered as x^* .

$$x^* = \frac{\int x \cdot \mu_{A_k}(x) dx}{\int \mu_{A_k}(x) dx}$$

7. First (or last) of maxima

In this method, the first element, say x_α , or the last element, say x_ω , is chosen as x^* .

$$x_\alpha = \min(\{x \mid \mu_A(x) = h(x)\})$$

$$x_\omega = \max(\{x \mid \mu_A(x) = h(x)\})$$